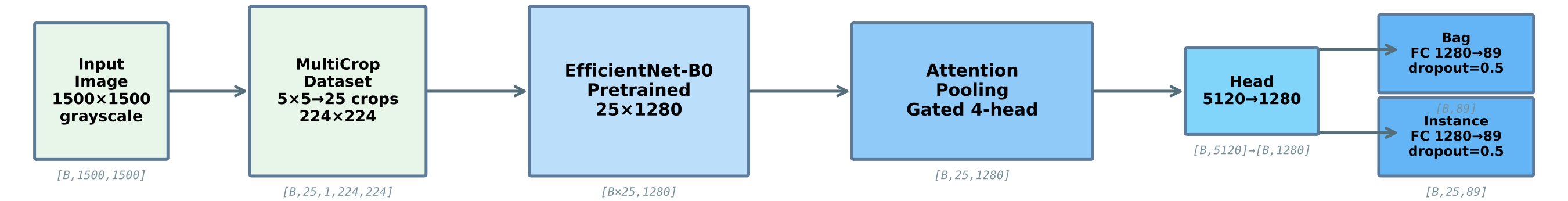
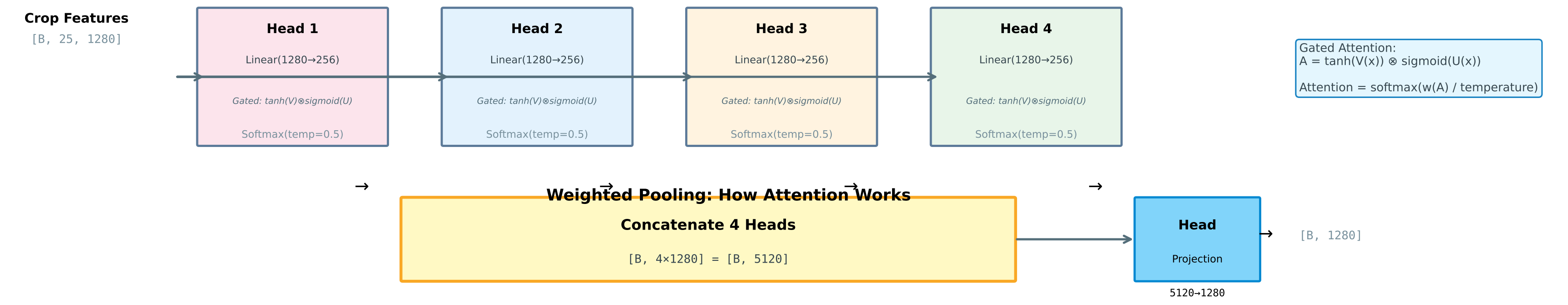
MIL Architecture - Final Mutant Model (Complete Pipeline)


Input Image 1500×1500 grayscale
[B,1500,1500]
MultiCrop Dataset 5×5→25 crops 224×224
[B,25,1,224,224]
EfficientNet-B0 Pretrained 25×1280
[B×25,1280]
Attention Pooling Gated 4-head
[B,25,1280]
Head 5120→1280
[B,5120]→[B,1280]
Bag FC 1280→89 dropout=0.5
[B,89]
Instance FC 1280→89 dropout=0.5
[B,25,89]

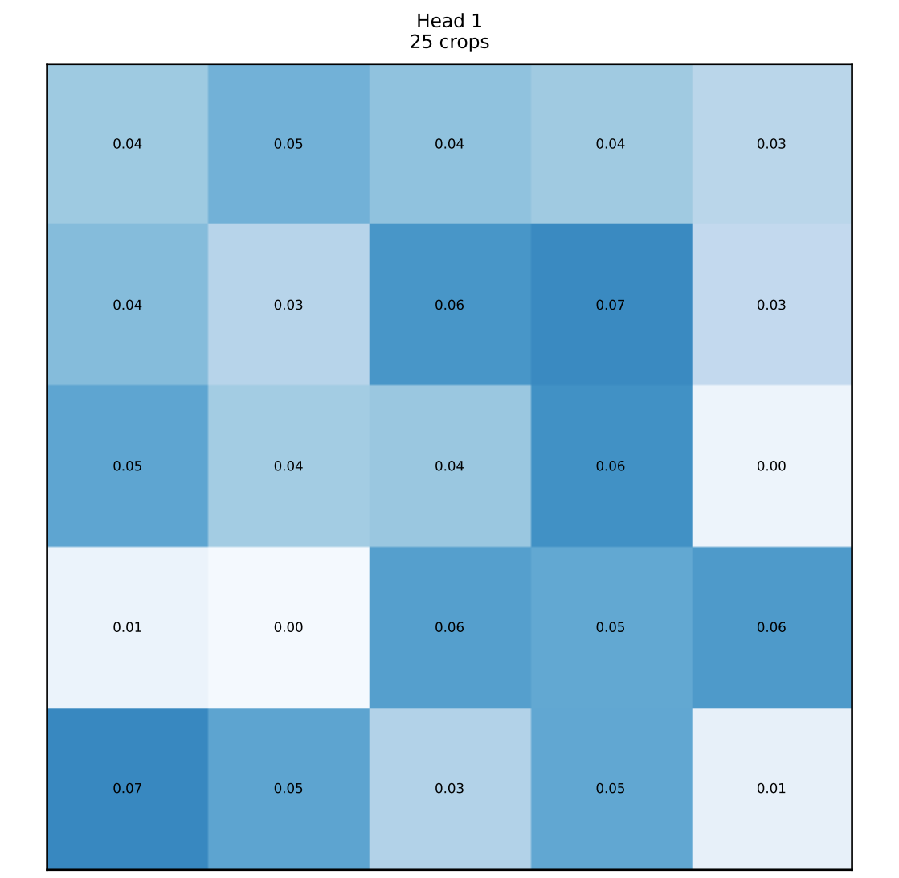
Gated Multi-Head Attention Pooling (Detailed)


Crop Features
[B, 25, 1280]
Head 1
Linear(1280→256)
Gated: tanh(V)⊗sigmoid(U)
Softmax(temp=0.5)
Head 2
Linear(1280→256)
Gated: tanh(V)⊗sigmoid(U)
Softmax(temp=0.5)
Head 3
Linear(1280→256)
Gated: tanh(V)⊗sigmoid(U)
Softmax(temp=0.5)
Head 4
Linear(1280→256)
Gated: tanh(V)⊗sigmoid(U)
Softmax(temp=0.5)
Weighted Pooling: How Attention Works
Concatenate 4 Heads
[B, 4×1280] = [B, 5120]
Head
Projection
5120→1280
[B, 1280]

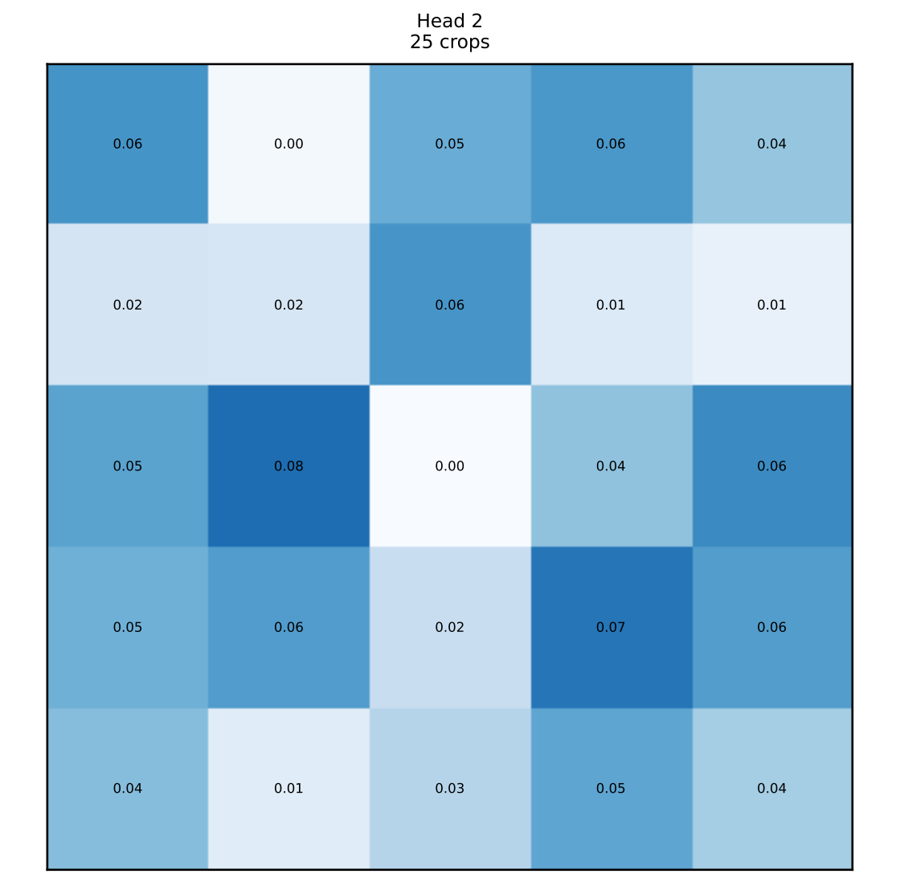
Gated Attention:
A = tanh(V(x)) ⊗ sigmoid(U(x))

Attention = softmax(w(A) / temperature)

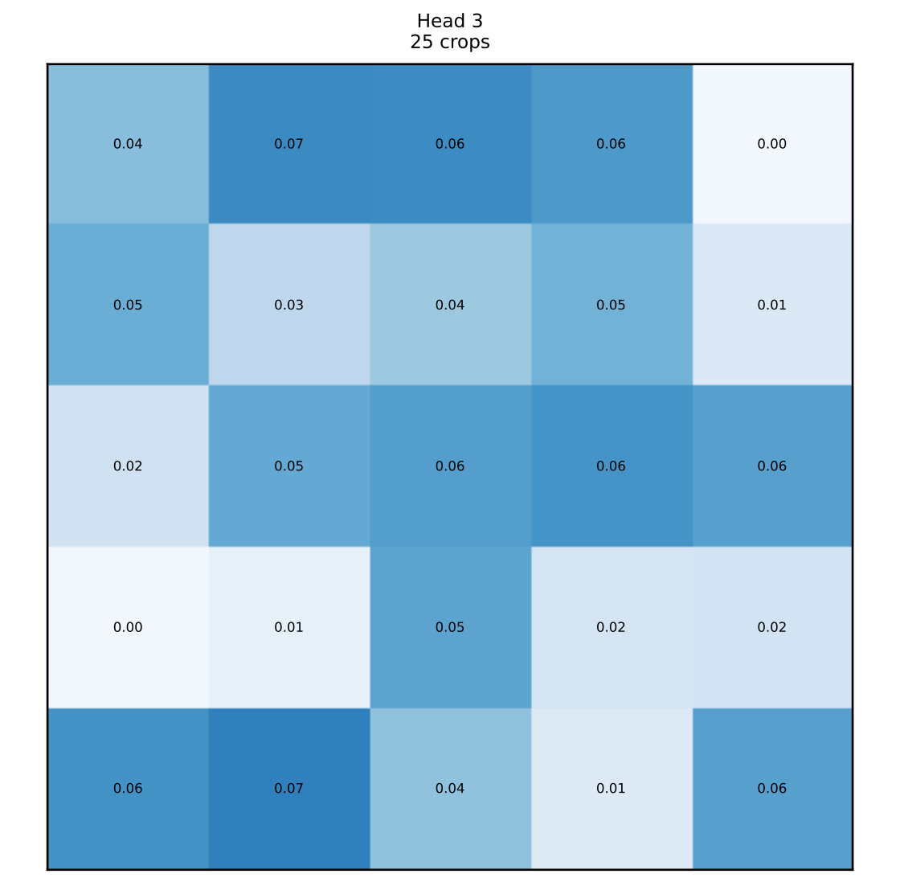
Attention Weights per Head:


Head 1
25 crops

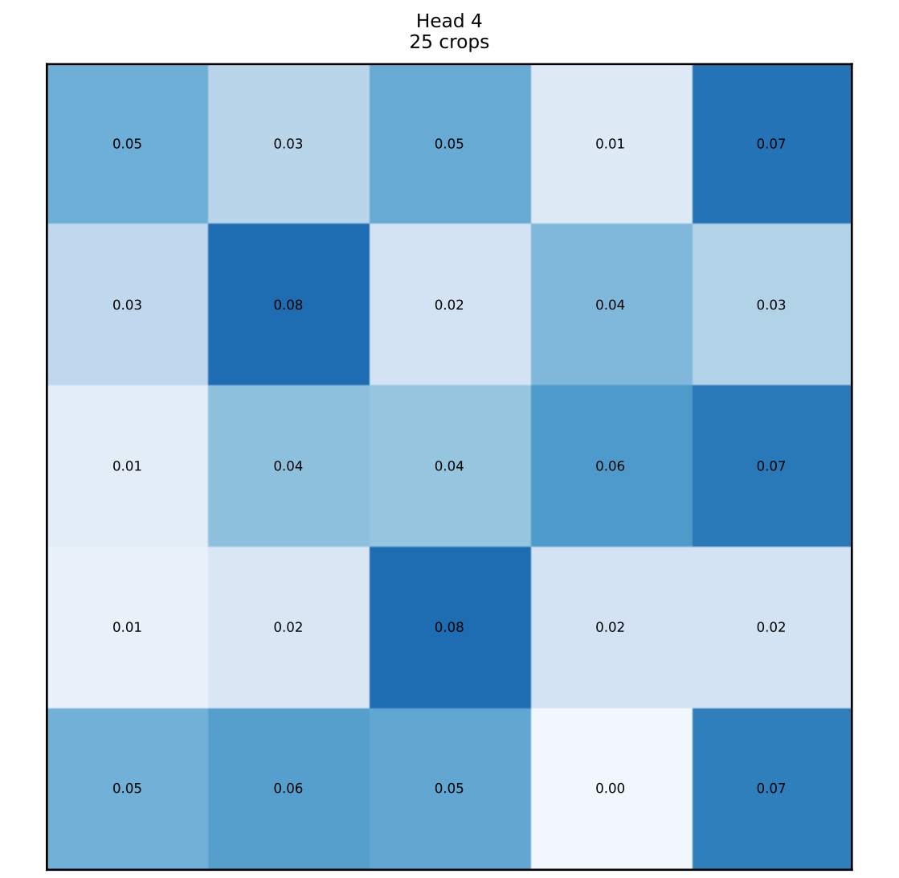
0.04	0.05	0.04	0.04	0.03
0.04	0.03	0.06	0.07	0.03
0.05	0.04	0.04	0.06	0.00
0.01	0.00	0.06	0.05	0.06
0.07	0.05	0.03	0.05	0.01


Head 2
25 crops

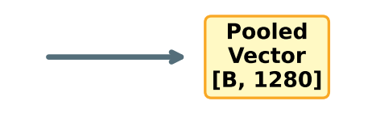
0.06	0.00	0.05	0.06	0.04
0.02	0.02	0.06	0.01	0.01
0.05	0.08	0.00	0.04	0.06
0.05	0.06	0.02	0.07	0.06
0.04	0.01	0.03	0.05	0.04


Head 3
25 crops

0.04	0.07	0.06	0.06	0.00
0.05	0.03	0.04	0.05	0.01
0.02	0.05	0.06	0.06	0.06
0.00	0.01	0.05	0.02	0.02
0.06	0.07	0.04	0.01	0.06


Head 4
25 crops

0.05	0.03	0.05	0.01	0.07
0.03	0.08	0.02	0.04	0.03
0.01	0.04	0.04	0.06	0.07
0.01	0.02	0.08	0.02	0.02
0.05	0.06	0.05	0.00	0.07


Pooled Vector [B, 1280]

pooled = Σ(attention_weights × features) for each head, then concatenate all heads